

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	03 June 2026
Team ID	2023011032040
Project Name	AI-Based Threat Intelligence Platform
Maximum Marks	4 Marks

Technical Architecture:

Based on the guidelines, here is the conceptual map for your Architecture Diagram. You can draw this using tools like Lucidchart or draw.io:

- 1. Processes (Application Logic): FastApi Backend, AI/ML Correlation Engine, Data Ingestion Pipeline.
- 2. Infrastructural Demarcation: Cloud-native deployment (AWS) utilizing Kubernetes for container orchestration.
- 3. External Interfaces: Third-party APIs (VirusTotal, AlienVault OTX) for external threat intelligence.
- 4. Data Storage Components: Amazon S3 (Raw Data Lake), PostgreSQL (Structured Relational Data), MongoDB (Unstructured Logs).
- 5. Machine Learning Interface: TensorFlow/PyTorch serving API for real-time anomaly detection.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How user interacts with application (Security Dashboard, Alerts UI)	React.js, Tailwind CSS, D3.js (for visualizations)
2	Application Logic-1	Core backend logic and API endpoints	Python / FastAPI
3	Application Logic-2	High-throughput data ingestion and stream processing	Apache Kafka, Logstash
4	Application Logic-3	Asynchronous task processing (alerting, batch jobs)	Celery, Redis

5	Database	Relational data (Users, Configs) & Unstructured data (Logs)	PostgreSQL, MongoDB
6	Cloud Database	Managed database services on the cloud	Amazon RDS, Amazon DocumentDB
7	File Storage	Raw telemetry and log storage (Data Lake)	Amazon S3
8	External API-1	External Threat Intelligence Feeds	VirusTotal API, AlienVault OTX
9	External API-2	Alerting and Notifications	Twilio API (SMS), SendGrid (Email)
10	Machine Learning Model	Anomaly detection and behavioral analysis	TensorFlow, Scikit-learn (Isolation Forest)
11	Infrastructure	Application Deployment on Cloud Environment	AWS (Amazon EKS for Kubernetes, Docker)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	List of open-source frameworks utilized in the platform	React, FastAPI, TensorFlow, ELK Stack (Elasticsearch, Logstash, Kibana)
2	Security Implementations	Security and access controls implemented	OAuth 2.0 / JWT for IAM, AES-256 (Data at rest), TLS 1.3 (Data in transit), AWS WAF
3	Scalable Architecture	Justification of architecture scalability	Microservices architecture utilizing Docker and Kubernetes (K8s) for horizontal pod auto-scaling
4	Availability	Justification of high availability	AWS Application Load Balancer (ALB), Multi-AZ (Availability Zone) deployments
5	Performance	Design considerations for performance	Amazon CloudFront (CDN) for UI, Redis for query/session caching, Kafka for

			decoupled high-speed data streams
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References:

- <https://aws.amazon.com/architecture/security-identity-compliance/>
- <https://reactjs.org/docs/getting-started.html>
- <https://fastapi.tiangolo.com/>
- <https://www.tensorflow.org/tfx/guide/serving>
- <https://kafka.apache.org/documentation/>